

AI Legal Aid Multi-Agent System IBM watsonx Hackathon 2026

1. Project Title

AI Legal Aid Multi-Agent System using IBM watsonx

2. Problem Statement

Problem Statement No. 3 - AI Legal Aid Multi-Agent System

Access to legal services is often expensive, time-consuming, and difficult for individuals and small businesses. Legal documents contain complex terminology, making it challenging for non-lawyers to understand contracts, rights, compliance requirements, and legal procedures.

This project provides an AI-powered legal assistant using IBM watsonx that combines Retrieval-Augmented Generation (RAG), document intelligence, and multiple AI agents to deliver understandable, accurate, and context-aware legal assistance.

3. Abstract

The AI Legal Aid Multi-Agent System is an intelligent legal assistant that helps users understand legal matters through natural conversations, document analysis, legal research, compliance verification, and contract review.

Unlike traditional chatbots, this system uses multiple specialized AI agents coordinated through IBM watsonx to solve different legal tasks collaboratively.

Users can upload contracts, ask legal questions, receive document summaries, identify legal risks, retrieve relevant laws and case precedents using Retrieval-Augmented Generation (RAG), and generate professional legal reports.

The system simplifies legal information into easy-to-understand language while always informing users that it is not a replacement for professional legal advice.

4. Objectives

The project aims to:

- * Provide affordable AI-powered legal assistance.
- * Simplify legal language into understandable explanations.
- * Analyze legal documents automatically.

- * Detect contractual risks.
- * Verify compliance with legal regulations.
- * Retrieve relevant legal information using RAG.
- * Generate structured legal reports.
- * Improve access to justice for citizens and small businesses.

5. Proposed Solution

The proposed solution is a Multi-Agent AI system where each intelligent agent specializes in a specific legal task.

Instead of one AI handling everything, multiple agents collaborate to produce accurate and explainable results.

The workflow includes:

User Login

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Dashboard

?

New Consultation

?

Intake Interview

?

Document Upload

?

Document Analysis

?

Legal Research

?

Compliance Checking

?

Contract Review

?

Legal Advice

?

Final Report Generation

6. System Architecture

User

?

Authentication

?

User Dashboard

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???

New Consultation Upload File Search Laws

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Intake Agent Document Parser RAG Agent

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AI Coordinator Agent

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Contract Compliance Case Research

Reviewer Agent Agent

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Legal Advice Agent

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?

Report Generator Agent

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?

Download PDF Report

7. Technologies Used

Frontend

- * React.js

- * HTML5

- * CSS3

- * Tailwind CSS

- * JavaScript

Backend

- * Python

- * Flask

- * FastAPI (optional)

AI

- * IBM watsonx.ai

- * Granite Foundation Models

- * LangChain

- * CrewAI / LangGraph

Database

- * PostgreSQL

- * SQLite (Development)

Vector Database

- * FAISS

- * ChromaDB

Authentication

- * JWT
- * Google OAuth
- * IBM Cloud Authentication

File Processing

- * PyMuPDF
 - * pdfplumber
 - * OCR
 - * Tesseract
- ### Deployment

- * IBM Cloud
- * Docker
- * GitHub

8. Modules

Module 1 - Authentication

Features

- * Login
- * Registration
- * Google Login
- * IBM Login
- * Forgot Password
- * JWT Authentication

Roles

- * Citizen
- * Lawyer
- * Administrator

Module 2 - Dashboard

Displays

- * Recent Cases

- * Notifications
- * Uploaded Documents
- * Saved Reports
- * Pending Consultations

Quick Actions

- * New Consultation
- * Upload Contract
- * Search Laws

Module 3 - Intake Interview Agent

Collects

- * Case Type
- * Country
- * State
- * Language
- * Description
- * Urgency

Dynamic questions based on selected legal category.

Module 4 - Document Upload

Supported formats

- * PDF
- * DOCX
- * TXT
- * JPG
- * PNG

Supported documents

- * Contracts
- * Agreements
- * Notices
- * Payslips
- * Offer Letters
- * Legal Notices

Module 5 - Document Parsing Agent

Extracts

- * Parties
- * Salary
- * Dates
- * Clauses
- * NDA
- * Arbitration
- * Confidentiality
- * Termination
- * Obligations

Provides

- * Summary
- * Highlighted Clauses
- * Risk Score

Module 6 - Contract Review Agent

Detects

- * Missing clauses
- * Legal inconsistencies
- * Unlimited liability
- * Missing termination clauses
- * IP ownership issues

Outputs

- * Severity
- * Recommendation
- * Explanation

Module 7 - Compliance Checker Agent

Checks

- * Labour Law
- * Consumer Law
- * Data Privacy
- * Employment Regulations

Output

PASS

FAIL

Warnings

Recommendations

Module 8 - Legal Research Agent

Uses Retrieval-Augmented Generation.

Retrieves

- * Constitution
- * Acts
- * Rules
- * Regulations
- * Supreme Court Judgments
- * High Court Judgments

Provides

- * Summary
 - * Citation
 - * Relevant Sections
- Module 9 - Conversational Legal Assistant

Allows users to ask legal questions.

Example

Can my employer terminate me without notice?

Provides

- * Simple explanation
 - * Relevant laws
 - * Legal references
 - * Disclaimer
- Module 10 - Report Generator

Generates

- * Client Summary
- * Facts
- * Legal Issues
- * Applicable Laws
- * Related Cases
- * Compliance Results
- * Contract Risks
- * Recommendations
- * Disclaimer

Export

- * PDF
- * DOCX

Module 11 - History

Stores

- * Previous Consultations
- * Uploaded Files
- * Generated Reports

Allows

- * Search
- * Delete
- * Reopen

Module 12 - Admin Dashboard

Manage

- * Users
- * AI Usage
- * Legal Database
- * RAG Documents
- * Analytics

9. Multi-Agent Design

Intake Agent

Collects user information.

Document Parser Agent

Reads uploaded legal documents.

Contract Reviewer Agent

Analyzes legal clauses and risks.

Compliance Checker Agent

Verifies legal compliance.

Legal Research Agent

Searches legal knowledge base using RAG.

Q&A Agent

Answers user-specific legal questions.

Legal Advice Agent

Provides understandable legal guidance.

Report Generator Agent

Creates downloadable reports.

10. Workflow

Login

?

Dashboard

?

New Consultation

?

Select Legal Category

?

Guided Intake Questions

?

Upload Documents

?

Document Parsing

?

Contract Review

?

Compliance Check

?

Legal Research

?

AI Legal Advice

?

Generate Report

?

Download PDF

...

11. IBM watsonx Integration

The application uses IBM watsonx Granite Foundation Models for:

* Natural language understanding

- * Legal reasoning
- * Conversational AI
- * Document summarization
- * Question answering
- * Report generation

Retrieval-Augmented Generation is implemented using a vector database containing legal documents, acts, and judgments.

12. Features

- ? Multi-Agent Architecture
- ? RAG-based Legal Search
- ? Contract Analysis
- ? Compliance Verification
- ? Case Research
- ? Legal Chatbot
- ? AI Report Generation
- ? Document Upload
- ? Risk Detection
- ? Secure Authentication
- ? User Dashboard
- ? Admin Panel
- ? History Tracking
- ? Download Reports
- ? Responsive UI
- ? IBM watsonx Integration

13. Future Enhancements

- * Voice-based legal assistant
- * Multilingual support
- * AI-generated legal drafts
- * Real-time lawyer consultation
- * Mobile application
- * Court hearing reminders
- * E-signature integration
- * AI confidence score
- * OCR for handwritten legal documents
- * Blockchain-based document verification

14. Advantages

- * Affordable legal assistance
- * Available 24/7
- * Faster legal research
- * Simplified legal language
- * Accurate document analysis
- * Automated compliance checking
- * Better access to justice
- * Time-saving
- * User-friendly interface

15. Limitations

- * Not a substitute for professional legal advice.
- * Depends on the quality and coverage of the legal knowledge base.
- * Jurisdiction-specific laws require regular updates.
- * Complex litigation still requires human legal experts.

16. Expected Outcomes

- * Faster access to legal information
- * Reduced legal consultation time
- * Improved understanding of contracts and legal rights
- * Better compliance awareness
- * Automated legal document analysis
- * Structured and downloadable legal reports

17. Testing Scenarios

Test Case	Expected Result
User Login	Successful authentication
Upload Contract	Document parsed successfully
Ask Legal Question	AI provides response with citations
Contract Review	Risks highlighted with recommendations
Compliance Check	PASS/FAIL with explanations
Legal Research	Relevant laws and case summaries retrieved
Generate Report	PDF/Word report generated
Search History	Previous consultations displayed
Admin Login	Admin dashboard accessible
Logout	Secure session termination

18. Conclusion

The "AI Legal Aid Multi-Agent System" demonstrates how IBM watsonx and specialized AI agents can make legal services more accessible, efficient, and understandable. By integrating Retrieval-Augmented Generation, document analysis, compliance verification, contract review, and conversational AI into a single platform, the system empowers individuals and small businesses to better understand legal issues while maintaining transparency through citations and appropriate legal disclaimers. This solution addresses the hackathon challenge by delivering a scalable, production-ready legal assistance platform that is user-friendly, secure, and extensible for future legal domains.